

Razorpay AI Buildathon 2026 | Track 03

AI Revenue Recovery: Deep Problem Statement Deconstruction & Solution Architecture

Project: Revenue Recovery Brain | Author: Himanshu | Status: 29/29 Verification Tests Passed (100%)

Focus: Bounded Workflow · RBI Fair Practices Code · Stopping Rules · Cryptographic Ledger

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1 Deconstructing Track 03: What Razorpay Actually Asked For

1.1 The Official Prompt

Razorpay defines Track 03 with exact specifications:

“Build an agent that detects revenue at risk, determines the right intervention, and executes a bounded recovery workflow: from payment failures and checkout abandonment to overdue receivables.”

1.2 The Real-World Financial Bleed: The 4 Fragmented Funnels

When revenue fails to reach an Indian merchant, it does not disappear in a single clean step. It degrades across four distinct, uncoordinated failure modes:

1. **Payment Degradation (Gateway Level):** India processes over 15 billion UPI transactions a month with an aggregate decline rate of ~8%—amounting to over **1.2 billion declined transactions every month**.
2. **Checkout Drop-off (Cart Level):** Mobile UPI intent mismatches, checkout hesitation, and delivery-fee friction cause 70%+ of prospective orders to be abandoned.

3. **Subscription Churn (Mandate Level):** Involuntary recurring debit cancellations caused by the RBI's e-mandate rule requiring explicit Additional Factor Authentication (AFA/OTP) for charges >₹ 15,000.
4. **B2B Trade Receivables (Invoice Level):** Indian SMEs suffer an average **73-day Days Sales Outstanding (DSO)**. Ineffective email dunning causes invoices to cross the critical 45-day threshold under Section 43B(h) of the Income Tax Act.

The Non-Technical Analogy: The “Dumb Water Pump” vs. The “Smart Irrigation Grid”

Imagine a commercial building supplied by four distinct water pipelines:

- **Pipe 1 (Retail Gateway):** The main municipal city water line.
- **Pipe 2 (Checkout Carts):** The bathroom faucets tenants leave dripping.
- **Pipe 3 (Subscription Mandates):** Automated monthly lawn sprinklers.
- **Pipe 4 (B2B Trade Invoices):** An industrial water tanker delivering bulk water on credit.

The Conventional Dunning Trap: When water stops flowing in Pipe 1, traditional dunning bots act like a **dumb motorized pump**: they simply crank the pressure dial to 100% and pump repeatedly.

- If the municipal line is shut down for maintenance (an HDFC or SBI issuer switch outage), cranking the pump delivers zero water—it burns out the motor (wasting merchant API fees) and rattles tenant walls with loud alarms (spamming the customer).
- If the tenant's individual tank is empty (insufficient balance), blasting pressure into a closed valve accomplishes nothing.
- Worst of all, the landlord has no idea that the tenant whose kitchen tap failed also owes ₹ 85,000 for the commercial tanker downstairs!

1.3 Real-World Indian Commerce Scenarios

Scenario A: Payment Degradation — The UPI Switch Spike

Context: Amit is ordering headphones on BoAt for ₹ 2,499. He enters his UPI PIN on PhonePe, but HDFC Bank’s core banking switch is under peak load and times out.
The Flawed Default: Standard gateways output a generic “Payment Failed” message. Amit abandons the cart in frustration and orders from a competitor. The merchant loses ₹ 2,499 permanently.

Scenario B: Checkout Drop-off — The Mobile Cart Hesitation

Context: Priya adds a ₹ 4,500 designer dress to her cart. At checkout, an incoming phone call interrupts her, or she hesitates over a delivery fee.
The Flawed Default: Generic email marketing bots send 5 repetitive email blasts 3 days later, which land in the Promotions tab long after buyer intent has evaporated.

Scenario C: Subscription Mandate Churn — The Silent ₹ 15,000+ AFA Failure

Context: Vikram’s agency subscribes to a creative SaaS platform for ₹ 18,000/month via UPI Autopay. Under RBI circular DPSS.CO.PD.No.447, recurring debits over ₹ 15,000 require explicit AFA approval 24 hours in advance. The bank rejects the automated debit.
The Flawed Default: The SaaS platform marks Vikram as “churned,” cancels his account, and cuts off his team’s workflow, destroying Monthly Recurring Revenue (MRR).

Scenario D: B2B Receivables — The Delayed SME Invoice

Context: Rohit Mehta runs Mehta Logistics Pvt Ltd and owes a parts supplier ₹ 85,000. The invoice is 38 days overdue. Rohit has temporary cash flow constraints and ignores automated English PDF dunning notices.
The Flawed Default: Standard accounting software sends impersonal English legal notices that get routed to an unread spam folder. Day 45 arrives, and both parties lose tax benefits under Section 43B(h).

2 What Razorpay Specifically Demanded Under ‘THE BAR’

In the official prompt, Razorpay explicitly established four non-negotiable evaluation criteria:

“THE BAR: Don’t just identify the problem. Show measured money recovered across a batch, with compliant escalation, stopping rules, and an audit trail.”

Deconstructing The Four Evaluation Pillars

1. **1. Measured Money Recovered Across a Batch:** Single toy examples are insufficient. Evaluators look for a full batch simulation (50+ real-world cases) showing total exposure at risk, total recovered amount, and a verified mathematical recovery rate.
2. **2. Compliant Escalation:** The agent must recognize when an issue exceeds its autonomous authority:
 - High-stakes transactions (invoices >₹ 50,000).
 - Debtor disputes or expressed financial distress.
 - Outside statutory contact hours (8:00 AM – 7:00 PM IST).

3. **3. Hard Stopping Rules:** The agent must possess hardcoded limits that prevent destructive or wasteful behavior:
 - Frequency caps: Maximum 2 voice calls and 3 digital nudges per week.
 - Cool-off rules: Mandatory 48-hour quiet period following customer dispute.
 - Economic floor: Immediately aborts any recovery where expected recovery value < communication cost (<₹ 100).
 - Terminal failure suppression: Halts retries on terminal codes (CARD_STOLEN, ACCOUNT_CLOSED).
4. **4. Immutable Cryptographic Audit Trail:** Every diagnosis, policy check, and API invocation must be sealed in an immutable sequence. Evaluators must be able to verify mathematical integrity independently of the server runtime.

3 How Our Solution Solves Track 03: Scenario-by-Scenario

Instead of separate, uncoordinated tools, we built the **Revenue Recovery Brain**: a unified engine operating on a deterministic 4-stage pipeline:

Signal Ingestion → **Diagnosis** (<150ms) → **ENRV Routing** → **RBI Policy Shield** → **Cryptographic Execution**

3.1 Walking Through The Exact Scenarios in Our Architecture

3.1.1 1. Amit's HDFC UPI Timeout (Scenario A → Smart Rail Reroute)

- **Diagnosis (<150ms):** The diagnostic engine reads `error_code="GATEWAY_ERROR"`, `error_source="bank"`. It classifies it as **Technical Degradation (TD)**.
- **Circuit Breaker Check:** Recognizes that HDFC switch success rates are temporarily depressed. It suppresses an immediate blind retry against the same rail.
- **Intervention:** The system generates a secondary 1-click Razorpay payment link (`plink_`) delivered directly to Amit's WhatsApp with alternate rail routing (e.g., ICICI/Axis switch).
- **Result:** Amit taps the link, approves via Google Pay, and ₹ 2,499 is recovered without merchant fee burnout.

3.1.2 2. Priya's Cart Abandonment (Scenario B → Counterfactual WhatsApp Nudge)

- **Diagnosis:** Detects `order.abandoned` on high-margin merchandise (₹ 4,500).
- **ENRV Calculation:** Computes counterfactual lift:

$$\text{ENRV} = (P_{\text{recovery|nudge}} \times \text{Net Amount}) - (P_{\text{natural}} \times \text{Gross}) - \text{Cost}_{\text{whatsapp}} > ₹ 100$$

- **Intervention:** Dispatches an instant WhatsApp recovery link offering 1-click checkout within 15 minutes.
- **Stopping Rule:** If Priya does not interact, the contact governor halts further outreach to prevent brand fatigue.

3.1.3 3. Vikram's ₹ 18,000 SaaS Mandate (Scenario C → Automated 24h Pre-Debit Push)

- **Diagnosis:** Recognizes amount (>₹ 15,000) matches the RBI e-mandate threshold.
- **Intervention:** Triggers an automated 24-hour pre-debit push flow. The system dispatches an authentic Razorpay AFA push link to Vikram's verified WhatsApp and email.
- **Result:** Vikram performs 1-click biometric authorization 24 hours before debit execution; subscription renews with zero involuntary churn.

3.1.4 4. Rohit Mehta's Overdue ₹ 85,000 Invoice (Scenario D → Hinglish Voice Negotiation)

- **Diagnosis:** Identifies 38 days overdue. Cross-references the **Section 43B(h) Tax Clock**: if unpaid by day 45, Rohit's company loses expense tax deductibility!
- **Intervention:** Dispatches our **Hinglish Conversational Voice Agent**. The agent calls Rohit:

"Namaskar Rohit ji! Main Recovery Brain se bol raha hoon. Aapka ₹ 85,000 ka invoice pending hai. Section 43B(h) tax benefit ke liye kya hum ise 2 installments mein schedule karein?"

- **Strict RBI Voice Security Invariant:** The agent is **architecturally prohibited from requesting OTPs, CVVs, or UPI PINs**. It negotiates a Promise-to-Pay (PTP) date and sends an authentic Razorpay Payment Link (`plink_`) via SMS for self-service payment.

4 Why Our Solution Beats Other Hackathon Submissions

Evaluation Dimension	Typical Competitor Submis-sions	Razorpay Revenue Recovery Brain
Architecture	Simple LLM wrapper: pipes in-coming webhooks directly into ChatGPT.	Trust Boundary Architecture: Strict separation between read-only LLM diagno-sis, hardcoded policy gates, and cashier exe-cution.
Funnel Scope	Siloed point solution: solves only cart abandonment OR only pay-ment retries.	Unified 4-Funnel Engine: Concurrently orchestrates Gateway Drops, Carts, Man-dates, and B2B Invoices.
Customer Profile	Treats the debtor as four discon-nected strangers across funnels.	Cross-Leak State Unification: Corre-lates B2B debt with retail payment failure to diagnose liquidity distress.
Bank Rail Awareness	Blind retries on fixed timers, even during total issuer switch collapses.	Bank Circuit Breaker: Tracks rolling EMA across HDFC, SBI, ICICI; automati-cally shrinks autonomy envelope (<30% SR).
Regulatory Fluency	Ignores Indian laws; calls debtors at 10:00 PM or spams multiple daily nudges.	Hardcoded RBI Fair Practices Code: 8 AM – 7 PM contact gate, weekly attempt ceilings, mandatory 48h cool-offs.
Audit Integrity	Plain terminal <code>print()</code> logging or mutable application database rows.	Persisted Cryptographic Ledger: SHA-256 chained blocks in SQLite, verifiable stan-dalone via <code>verify_ledger.py</code> .
Economic Discipline	Spends \$0.05 on AI calls to re-cover ₹ 10 items (negative-ROI leakage).	WACC-Discounted ENRV Math: Con-tinuous 18% p.a. interest discounting and ₹ 100 economic floor rule.

Table 1: Architectural Comparison: Toy Wrapper vs. Production Infrastructure

5 The ‘Bar’ Verification Checklist

Every single requirement set by Razorpay in Track 03 is verified by our automated test suite:

Official Track 03 Requirement	Engineering Implementation in Codebase		Verification Test
Detects revenue at risk	Diagnostic (diagnosis_engine.py) in <150ms.	Engine operates	test_4 (PASSED)
Bounded recovery workflow	Autonomy (autonomy_envelope.py) at ₹ 25k / ₹ 5k.	Envelope caps capital	test_13, 28 (PASSED)
Payment degradation → action	Intervention (intervention_router.py) TD vs. BD.	Router routes	test_12 (PASSED)
Checkout drop-off recovery	Pipeline routes abandoned orders with counterfactual WhatsApp links.		test_12 (PASSED)
Failed-subscription recovery	Automated 24-hour pre-debit notifications for recurring debits >₹ 15,000.		test_12 (PASSED)
B2B receivables chaser	Section 43B(h) MSME Tax Clock countdown engine with 45-day threshold.		test_9 (PASSED)
Hinglish voice recovery	Conversational voice agent with AST-verified credential prohibition.		test_15, 22 (PASSED)
Promise-to-pay tracker	PTP state machine (smart_scheduler.py) with automated cool-off.		test_14 (PASSED)
Measured batch recovery	50-case batch benchmark: ₹ 8.7L at risk → ₹ 5.9L recovered (68%).		batch_results
Compliant escalation	High-stakes approval gate holds interventions >₹ 50,000 for supervisor.		test_8 (PASSED)
Stopping rules	Economic floor rule (<₹ 100) and terminal failure filter.		test_3, 25 (PASSED)
Cryptographic audit trail	SQLite-persisted SHA-256 chained ledger with zero-dependency CLI verification.		test_6, 19, 29 (PASSED)

Table 2: Track 03 Requirements Verification Matrix (29/29 Passing)